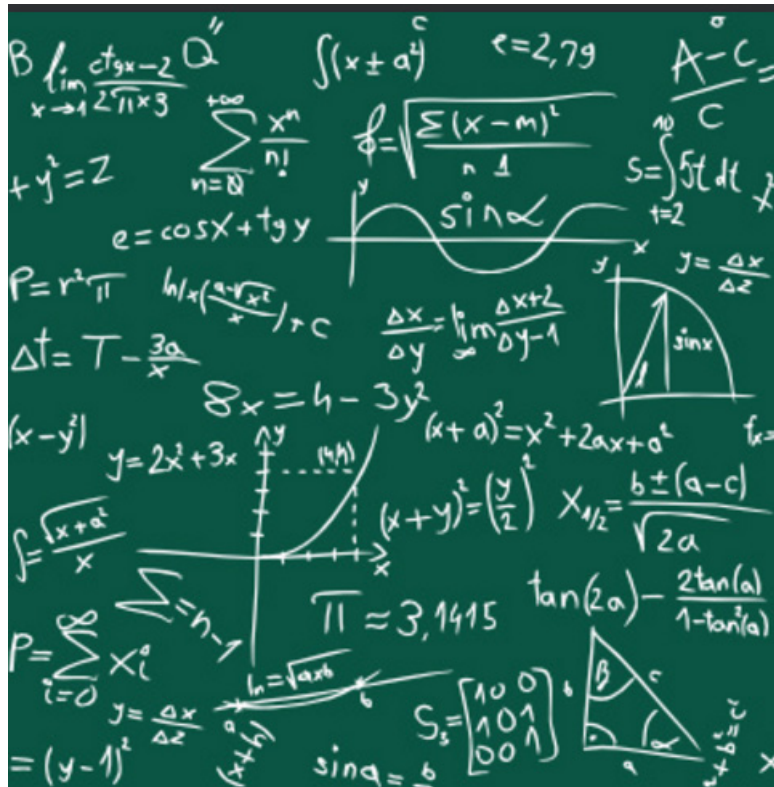


Exercises: Limits via asymptotic expansions



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

Exercise 1

Compute the following limits:

$$\lim_{x \rightarrow 0} \frac{6(x - \sin x) - x^3}{\sin x^5}$$

$$\lim_{x \rightarrow 0} \frac{\sinh^2 x + 2(1 - \cosh x)}{(1 - \cos x)^2}$$

$$\lim_{x \rightarrow 0} \frac{4(1 - \cos x)^2 - x^2 \sin^2 x}{\sin^2 x \log(1 + x^4)}$$

$$\lim_{x \rightarrow 0} \frac{1 + x \sin x - e^{x^2}}{x \sin(x^3)}$$

$$\lim_{x \rightarrow 0} \frac{x^2 \cos x - \sinh x^2 + \frac{1}{2}x^4}{x^2 - \arctan x^2}$$

$$\lim_{x \rightarrow 0} \frac{x \sinh x - 2 \cosh x + 2}{(e^{\sin x} - 1)^2}$$

$$\lim_{x \rightarrow 0} \frac{8\sqrt{1 + \sin x} - 8 - 4x + x^2}{(2e^x - 2 - 2x - x^2) \cosh^2 x}$$

$$\lim_{x \rightarrow 0} \frac{2 \log(1 + \sin x) - 2x + x^2}{(2e^x - 2 - 2x - x^2) \cos^2 x}$$

$$\lim_{x \rightarrow +\infty} x \left(\sqrt{4 + \frac{5}{x}} - 2 \cos \frac{1}{x} \right)$$

We determine the principal part of the Maclaurin expansion of the numerator:

$$6(x - \sin x) - x^3 = 6 \left(\frac{x^3}{6} - \frac{x^5}{120} + o(x^5) \right) - x^3 = -\frac{x^5}{20} + o(x^5).$$

The denominator is equivalent to x^5 :

$$\sin x^5 \sim x^5.$$

Hence the given limit is

$$\lim_{x \rightarrow 0} \frac{6(x - \sin x) - x^3}{\sin x^5} = \lim_{x \rightarrow 0} \frac{-\frac{x^5}{20}}{x^5} = -\frac{1}{20}$$

We determine the principal part of the Maclaurin expansion of the numerator:

$$\sinh^2 x + 2(1 - \cosh x) = \left(x + \frac{x^3}{6} + o(x^3)\right)^2 - x^2 - \frac{x^4}{12} + o(x^4) = \frac{x^4}{4} + o(x^4).$$

The denominator is equivalent to $x^4/4$:

$$(1 - \cos x)^2 \sim \frac{x^4}{4}.$$

Hence the given limit is

$$\lim_{x \rightarrow 0} \frac{\sinh^2 x + 2(1 - \cosh x)}{(1 - \cos x)^2} = \lim_{x \rightarrow 0} \frac{\frac{x^4}{4}}{\frac{x^4}{4}} = 1.$$

We determine the principal part of the Maclaurin expansion of the numerator:

$$\begin{aligned} 4(1 - \cos x)^2 - x^2 \sin^2 x &= 4\left(\frac{x^2}{2} - \frac{x^4}{24} + o(x^4)\right)^2 - x^2\left(x - \frac{x^3}{6} + o(x^3)\right)^2 \\ &= \frac{x^6}{6} + o(x^6). \end{aligned}$$

The denominator is equivalent to x^6 :

$$\sin^2 x \log(1 + x^4) \sim x^2 \cdot x^4 = x^6.$$

Hence the given limit is

$$\lim_{x \rightarrow 0} \frac{4(1 - \cos x)^2 - x^2 \sin^2 x}{\sin^2 x \log(1 + x^4)} = \lim_{x \rightarrow 0} \frac{\frac{x^6}{6}}{x^6} = \frac{1}{6}.$$

We determine the principal part of the Maclaurin expansion of the numerator:

$$\begin{aligned} 1 + x \sin x - e^{x^2} &= 1 + x \left(x - \frac{x^3}{6} + o(x^3) \right) - \left(1 + x^2 + \frac{x^4}{2} + o(x^4) \right) \\ &= -\frac{2}{3}x^4 + o(x^4). \end{aligned}$$

The denominator is equivalent to x^4 :

$$x \sin(x^3) \sim x \cdot x^3 = x^4.$$

Hence the given limit is

$$\lim_{x \rightarrow 0} \frac{1 + x \sin x - e^{x^2}}{x \sin(x^3)} = \lim_{x \rightarrow 0} \frac{-\frac{2}{3}x^4}{x^4} = -\frac{2}{3}.$$

We determine the principal part of the Maclaurin expansion of the numerator:

$$\begin{aligned} x^2 \cos x - \sinh x^2 + \frac{1}{2}x^4 \\ = x^2 \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4) \right) - \left(x^2 + \frac{x^6}{6} + o(x^6) \right) + \frac{1}{2}x^4 = -\frac{x^6}{8} + o(x^6). \end{aligned}$$

For the denominator, from the expansion $\arctan x = x - \frac{x^3}{3} + o(x^3)$, it follows that

$$x^2 - \arctan x^2 = \frac{x^6}{3} + o(x^6).$$

Hence the given limit is:

$$\lim_{x \rightarrow 0} \frac{x^2 \cos x - \sinh x^2 + \frac{1}{2}x^4}{x^2 - \arctan x^2} = \lim_{x \rightarrow 0} \frac{-\frac{x^6}{8}}{\frac{x^6}{3}} = -\frac{3}{8}.$$

We determine the principal part of the Maclaurin expansion of the numerator:

$$\begin{aligned} & x \sinh x - 2 \cosh x + 2 \\ &= x \left(x + \frac{x^3}{6} + o(x^3) \right) - 2 \left(1 + \frac{x^2}{2} + \frac{x^4}{24} + o(x^4) \right) + 2 = \frac{x^4}{12} + o(x^4). \end{aligned}$$

The denominator is equivalent to x^2 :

$$(e^{\sin x} - 1)^2 \sim (\sin x)^2 \sim x^2.$$

Hence the given limit is

$$\lim_{x \rightarrow 0} \frac{x \sinh x - 2 \cosh x + 2}{(e^{\sin x} - 1)^2} = \lim_{x \rightarrow 0} \frac{\frac{x^4}{12}}{x^2} = \lim_{x \rightarrow 0} \frac{x^2}{12} = 0.$$

We expand to order 3 the composite function $8\sqrt{1 + \sin x}$ with initial point $x = 0$. From $\sin x \sim x - x^3/6$ and $8\sqrt{1 + y} \sim 8 + 4y - y^2 + y^3/2$, by composition, we have

$$\begin{aligned} 8\sqrt{1 + \sin x} &= 8 + 4 \left(x - \frac{x^3}{6} \right) - \left(x - \frac{x^3}{6} \right)^2 + \frac{1}{2} \left(x - \frac{x^3}{6} \right)^3 + o(x^3) \\ &= 8 + 4x - x^2 - \frac{1}{6}x^3 + o(x^3). \end{aligned}$$

It follows that the numerator, in the given limit, is equivalent to $-x^3/6$.

In the denominator we have the factor $\cosh^2 x$ which converges to 1, while

$$2e^x - 2 - 2x - x^2 \sim x^3/3.$$

The given limit is

$$\lim_{x \rightarrow 0} \frac{8\sqrt{1 + \sin x} - 8 - 4x + x^2}{(2e^x - 2 - 2x - x^2) \cosh^2 x} = \lim_{x \rightarrow 0} \frac{-\frac{x^3}{6}}{\frac{x^3}{3}} = -\frac{1}{2}.$$

We expand to order 3 the composite function $2\log(1 + \sin x)$ with initial point $x = 0$. From $\sin x \sim x - x^3/6$ and $2\log(1 + y) \sim 2y - y^2 + 2y^3/3$, by composition, we have

$$\begin{aligned} 2\log(1 + \sin x) &= 2\left(x - \frac{x^3}{6}\right) - \left(x - \frac{x^3}{6}\right)^2 + \frac{2}{3}\left(x - \frac{x^3}{6}\right)^3 + o(x^3) \\ &= 2x - x^2 + \frac{1}{3}x^3 + o(x^3). \end{aligned}$$

It follows that the numerator, in the given limit, is equivalent to $x^3/3$. In the denominator we have the factor $\cos^2 x$ which converges to 1, while

$$2e^x - 2 - 2x - x^2 \sim x^3/3.$$

The given limit is

$$\lim_{x \rightarrow 0} \frac{2\log(1 + \sin x) - 2x + x^2}{(2e^x - 2 - 2x - x^2) \cos^2 x} = \lim_{x \rightarrow 0} \frac{\frac{1}{3}x^3}{\frac{1}{3}x^3} = 1.$$

$$\lim_{x \rightarrow +\infty} x \left(\sqrt{4 + \frac{5}{x}} - 2 \cos \frac{1}{x} \right) = \lim_{x \rightarrow +\infty} 2x \left(\sqrt{1 + \frac{5}{4x}} - \cos \frac{1}{x} \right)$$

From $\cos y = 1 + o(y)$ and $\sqrt{1 + y} = 1 + \frac{1}{2}y + o(y)$, by composition, we have

$$\begin{aligned} \lim_{x \rightarrow +\infty} 2x \left(\sqrt{1 + \frac{5}{4x}} - \cos \frac{1}{x} \right) &= \lim_{x \rightarrow +\infty} 2x \left(1 + \frac{1}{2} \cdot \frac{5}{4x} + o\left(\frac{1}{x}\right) - 1 + o\left(\frac{1}{x}\right) \right) \\ &= \lim_{x \rightarrow +\infty} 2x \left(\frac{5}{8x} + o\left(\frac{1}{x}\right) \right) \\ &= \frac{5}{4} + o(1) \\ &= \frac{5}{4} \end{aligned}$$